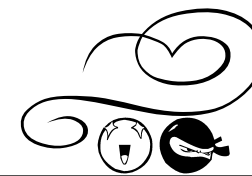


Binomials

Part A: Terms



Unit

NAME: _____

A1	BAULKHAM HILLS	☒	☐
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Consider the expansion $\left(\frac{x^3}{2} + \frac{a}{x}\right)^8$. If the constant term is 5103, determine the possible value(s) of a . 3

$$\left(\frac{x^3}{2} + \frac{a}{x}\right)^8; T_{k+1} = \binom{8}{k} \left(\frac{x^3}{2}\right)^{8-k} \left(\frac{a}{x}\right)^k$$

$$= \binom{8}{k} \frac{a^k}{2^{8-k}} x^{24-4k}$$

constant term has $x^0 \Rightarrow 24 - 4k = 0$
 $k = 6$

constant is $\binom{8}{6} \frac{a^6}{2^2}$, thus $\binom{8}{6} \frac{a^6}{2^2} = 5103$
 $7a^6 = 5103$
 $a^6 = 729$
 $a = \pm 3$

A2	BAULKHAM HILLS	☒	☐
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What is the coefficient of x^3 in the expansion of $(1 + x + x^2)^5$? 1

- (A) 10 (B) 20 (C) 30 (D) 40

The only terms above that will include x^3 in their expansions are

$$\binom{5}{4}(1+x)^4 x^2 \text{ and } \binom{5}{0}(1+x)^5$$

$$= 5x^2(1+x)^4 = (1+x)^5$$

Terms containing x^3 will be

$$5x^2 \cdot \binom{4}{1}x + \binom{5}{3}x^3$$

$$= 20x^3 + 10x^3$$

$$= 30x^3 \quad \therefore \text{Coefficient} = 30$$

A3	BAULKHAM HILLS	☒	☐
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If $(1 + kx)^n = 1 - 12x + 60x^2 - \dots$, find the values of k and n . 3

e) From $T_2: \binom{n}{1}(kx)^1 = -12x$
 $nkx = -12x$
 $nk = -12$ ①

From $T_3: \binom{n}{2}(kx)^2 = 60x^2$
 $\binom{n}{2}k^2 = 60$
 $\frac{n(n-1)}{2} \cdot k^2 = 60$
 $k^2 n(n-1) = 120$ ②

From ①, $k = \frac{-12}{n}$

Into ② $\left(\frac{-12}{n}\right)^2 \cdot n(n-1) = 120$
 $\frac{144}{n^2} \cdot n(n-1) = 120$
 $144n^2 - 144n = 120n^2$
 $24n^2 - 144n = 0$
 $24n(n-6) = 0$
 $n = 6$ or $n = 0$
 $\therefore$ Since $nk = -12$, $6k = -12$
 $k = -2$

A4	CHELTENHAM GIRLS	☒	☐
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Find the coefficient of the term containing x^3 in the expansion of $\left(2x + \frac{3}{x}\right)^7$ 3

$n=5 \rightarrow 7C5 (2^5) (3^2) = 6048$

A5	GOSFORD	☒	☐
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Find the coefficient of the term containing x^3 in the expansion of $\left(2x + \frac{3}{x}\right)^{11}$. 3

$$\left(2x + \frac{3}{x}\right)^{11} = {}^{11}C_0 (2x)^{11} \left(\frac{3}{x}\right)^0 + {}^{11}C_1 (2x)^{10} \left(\frac{3}{x}\right)^1 + \dots$$

Term containing x^3 is ${}^{11}C_7 (2x)^7 \left(\frac{3}{x}\right)^4$
 $= {}^{11}C_7 \cdot 2^7 x^7 \cdot 3^4 x^{-4}$
 $= {}^{11}C_7 \cdot 2^7 \cdot 3^4 x^3$
 $= 3421 + 40x^3$
 $\therefore$ Coefficient is 342140

A6	HURLSTONE	☒	☐
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Find the co-efficient of x in $\left(x + \frac{4}{x}\right)^5$ 2

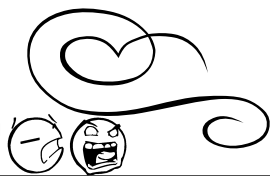
General Term = ${}^5C_r x^{5-r} \left(\frac{4}{x}\right)^r = {}^5C_r 4^r x^{5-2r}$
For $5-2r=1 \rightarrow r=2$
So, coefficient = ${}^5C_2 4^2 = 160$

A7	INTERNATIONAL GRAMMAR	☒	☐
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Find the coefficient of x^4 in $(x+2)^2(3+x^2)^5$ 3

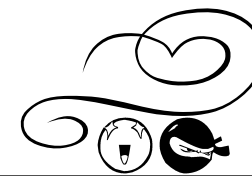
$$\Rightarrow (405x^2 \cdot x^2) + 4 \cdot 270x^4 \dots$$

$$= 1485x^4$$



Binomials

Part A: Terms



A8	INTERNATIONAL GRAMMAR	☒	☐
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What is the coefficient of x^7 in the expansion of $\left(x + \frac{4}{x}\right)^{13}$? 1

- A. 18 304 B. 45 760 C. 286 D. 64

A9	INTERNATIONAL GRAMM	☒	☐
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$$\begin{aligned} \binom{13}{r} x^{13-r} \left(\frac{4}{x}\right)^r & \quad 13-r-r=7 \\ 6=2r & \\ r=3 & \end{aligned}$$

$$\binom{13}{3} x^{10} \left(\frac{4}{x}\right)^3 = 18304x^7$$

Write the expansion of $(x - 2y)^4$.

$$\begin{aligned} & {}^4C_0 x^4 + {}^4C_1 x^3 (-2y) + {}^4C_2 x^2 (-2y)^2 + {}^4C_3 x (-2y)^3 + {}^4C_4 (-2y)^4 \\ & = x^4 - 8x^3 y + 24x^2 y^2 - 32xy^3 + 16y^4 \end{aligned}$$

A10	NEAP	☒	☐
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Find the coefficient of x^4 in the expansion of $\left(2 + \frac{x}{4}\right)^6$. 2

$$\begin{aligned} & A10 \\ & \left(2 + \frac{x}{4}\right)^6 : x^4 \\ & T_{k+1} = {}^6C_k 2^{6-k} \left(\frac{x}{4}\right)^k \\ & x^k = x^4 \\ & \therefore k=4 \\ & \therefore {}^6C_4 2^2 \left(\frac{x}{4}\right)^4 \\ & = 15 \times 4 \cdot \frac{x}{256} = \frac{15x}{64} \end{aligned}$$

A11	PEM	☒	☐
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Find the coefficient of x^2 in the expansion $\left(2x + \frac{1}{2x}\right)^8$ 3

$$\begin{aligned} T_{k+1} &= {}^8C_k (2x)^{8-k} \left(\frac{1}{2x}\right)^k \quad \text{for } x^2 \Rightarrow 8-2k=2 \\ &= {}^8C_k 2^{8-k} x^{8-k} \left(\frac{1}{2^k x^k}\right) \quad k=3 \\ &= {}^8C_k 2^{8-2k} x^{8-2k} \quad \text{the coefficient is } {}^8C_3 \times 2^2 = 224 \end{aligned}$$

A12	PEM	☒	☐
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Find the constant term in the expansion $\left(x + \frac{2}{x}\right)^6$. 3

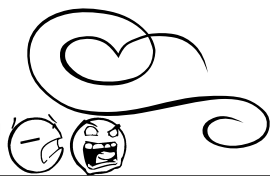
$$\begin{aligned} T_{k+1} &= {}^6C_k (x)^{6-k} \left(\frac{2}{x}\right)^k \quad \text{for constant term} \Rightarrow 6-2k=0 \\ &= {}^6C_k \times 2^k x^{6-k} \left(\frac{1}{x^k}\right) \quad k=3 \\ &= {}^6C_k \times 2^k x^{6-2k} \quad \text{the constant term is } {}^6C_3 \times 2^3 = 160 \end{aligned}$$

A13	PLC	☒	☐
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Expand $\left(x - \frac{2}{x}\right)^4$ and express your answer in its simplest form. 2

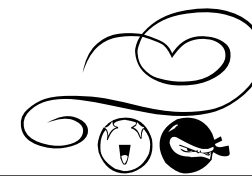
$$\begin{aligned} \left(x - \frac{2}{x}\right)^4 &= \binom{4}{0} x^4 (-2)^0 (x^{-1})^0 + \binom{4}{1} x^3 (-2)^1 (x^{-1})^1 + \binom{4}{2} x^2 (-2)^2 (x^{-1})^2 + \binom{4}{3} x^1 (-2)^3 (x^{-1})^3 + \binom{4}{4} x^0 (-2)^4 (x^{-1})^4 \\ &= \frac{16}{x^4} - \frac{32x}{x^2} + \frac{24x^2}{x^2} - \frac{8x^3}{x} + x^4 \\ &= \frac{16}{x^4} - \frac{32}{x} + 24 - 8x + x^4 \end{aligned}$$

no need for answer with common denominator



Binomials

Part A: Terms



A14	TOTAL EDUCATION	☒	☐
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What is the term independent of x in $(1 + \frac{3}{x})^2 (1 - 2x)^6$? 1

- A. 1 B. -72 C. 469 D. 540

$$\begin{aligned} (1 + \frac{3}{x})^2 &= 1 + \frac{6}{x} + \frac{9}{x^2} \quad \text{--- (1)} \\ (1 - 2x)^6 &= 1 - 6(2x) + 15(2x)^2 - 20(2x)^3 + \dots \\ &= 1 - 12x + 60x^2 - 160x^3 + \dots \quad \text{--- (2)} \end{aligned}$$

multiplying (1) $\times$ (2) leaves only the terms independent of x
thus $1 - (6 \times 12) + 9 \times 60$
 $= 469$

A15	TOTAL EDUCATION	☒	☐
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Consider the expansion of $(3x + \frac{c}{x})^8$ where $c > 0$.

- (i) Show that the general term of the expansion is $\binom{8}{i} 3^{8-i} c^i x^{-i}$. 1
(ii) Find the coefficient of x^4 . 1
(iii) Given that the coefficients of x^4 and x^6 are equal, find c . 2

$$\begin{aligned} \text{(i) General term of } (3x + \frac{c}{x})^8 &= \binom{8}{i} (3x)^{8-i} c^i x^{-i} \\ &= \binom{8}{i} 3^{8-i} c^i x^{8-2i} \\ &= \binom{8}{i} 3^{8-i} c^i x^{8-2i} \end{aligned}$$

(ii) Coeff of $x^4 \Rightarrow 8-2i=4 \Rightarrow i=2$
 $\therefore C_4 = \binom{8}{2} 3^6 c^2$
 $= 20412 c^2$

(iii) $C_4 = C_6 \Rightarrow 20412 c^2 = 17496 c$
 $\Rightarrow 20412 c = 17496$
 $\Rightarrow c = \frac{17496}{20412} = \frac{6}{7}$
NB: this works as $c > 0$

A16	TOTAL EDUCATION	☒	☐
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What is the constant term in the expansion of $(x - \frac{2}{x^3})^8$? 1

- A. 112 B. 120 C. 180 D. 224

3. General term in the expansion takes the form:
 $\binom{8}{r} (x)^{8-r} (-2x^{-3})^r = \binom{8}{r} (-2)^r (x)^{8-4r}$
For the constant term, we need to consider $8 - 4r = 0$
 $r = 2$
Therefore, for the constant term, $\binom{8}{2} (-2)^2 = 112$. Hence, correct option is A.

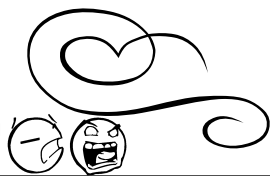
A17	TOTAL EDUCATION	☒	☐
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Given that the term independent of x in the expansion of $x^2 (2x^2 + \frac{k}{x^4})^8$ is -448 , find the exact value of k . 3

Sample answer:

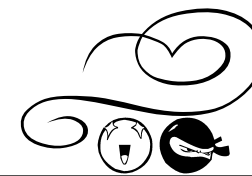
Rewrite $x^2 (2x^2 + \frac{k}{x^4})^8$ as:
 $x^2 (x^{-4} (2x^6 + k))^8 = x^2 \times x^{-32} (2x^6 + k)^8 = x^{-30} (2x^6 + k)^8$

Now we need the term with x^{30} from the expansion $(2x^6 + k)^8$. This will be the fourth term in the form ${}^8C_3 (2x^6)^5 (k)^3$. Consider the coefficient of this term,
 ${}^8C_3 (2)^5 (k)^3 = -448$
 $\therefore k^3 = \frac{-1}{4}$
 $\therefore k = \sqrt[3]{\frac{-1}{4}}$



Binomials

Part A: Terms



A18	ACE	☒	☐
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Find the term independent of x in the expansion of $\left(x - \frac{1}{2x^3}\right)^{20}$ 3

General term ${}^{20}C_k x^{20-k} \left(-\frac{1}{2x^3}\right)^k$
 $= {}^{20}C_k x^{20-k} \times x^{-3k} \times \left(-\frac{1}{2}\right)^k$
 independent of x means x^0
 $\therefore 20 - k - 3k = 0$
 $k = 5$
 $\therefore$ term is ${}^{20}C_5 \left(-\frac{1}{2}\right)^5 = -484.5$

A19	ACE	☒	☐
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Find the constant term in the expansion $\left(2x + \frac{3}{x^3}\right)^8$ 2

d) $\left(2x + \frac{3}{x^3}\right)^8$
 General term $= {}^8C_k (2x)^{8-k} (3x^{-3})^k$
 $= {}^8C_k 2^{8-k} x^{8-k} 3^k x^{-3k}$
 $= {}^8C_k 2^{8-k} 3^k x^{8-4k}$
 $\therefore 8 - 4k = 0$
 $k = 2$
 $\therefore$ Constant term $= {}^8C_2 2^6 3^2$
 $= 16128$ ✓

A20	BAULKHAM	☒	☐
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What is the term independent of x in the expansion of $(1+2x)^2 \left(2x + \frac{1}{x}\right)^{12}$ 1

- (A) $\binom{12}{6} \times 2^6$
 (B) $\binom{12}{6} \times 2^6 + \binom{12}{7} \times 2^6$
 (C) $\binom{12}{6} \times 2^6 + \binom{12}{7} \times 2^7$
 (D) $\binom{12}{6} \times 2^6 + \binom{12}{7} \times 2^9$

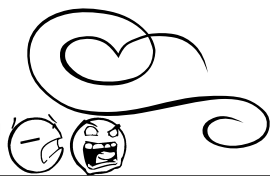
$(1+2x)^2 \left(2x + \frac{1}{x}\right)^{12}$
 $= (1+4x+4x^2) \left[(2x)^{12} + \binom{12}{1} (2x)^{11} \left(\frac{1}{x}\right) + \dots + \binom{12}{6} (2x)^6 \left(\frac{1}{x}\right)^6 + \dots + \binom{12}{7} (2x)^7 \left(\frac{1}{x}\right)^7 + \dots + \left(\frac{1}{x}\right)^{12} \right]$
 $= (1+4x+4x^2) \left[2^{12} x^{12} + \binom{12}{1} 2^{11} x^{11} + \dots + \binom{12}{6} 2^6 + \binom{12}{7} 2^7 \frac{1}{x} + \dots + \frac{1}{x^{12}} \right]$
 The term independent of x is $\binom{12}{6} 2^6 + 4 \times \binom{12}{7} 2^5 = \binom{12}{6} 2^6 + \binom{12}{7} 2^7$ C

A21	KWS	☒	☐
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What is the sixth term in the expansion of $(2x - 3y)^9$? 1

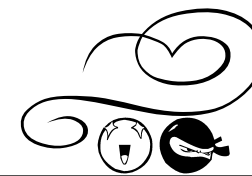
- (A) ${}^9C_3 \times 2^6 \times (-3)^3 x^6 y^3$
 (B) ${}^9C_4 \times 2^5 \times (-3)^4 x^5 y^4$
 (C) ${}^9C_5 \times 2^4 \times (-3)^5 x^4 y^5$
 (D) ${}^9C_6 \times 2^3 \times (-3)^6 x^3 y^6$

4. 9C_5 (C)



Binomials

Part A: Terms



A22	ABBOTSLEIGH	☒	☐
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Given $(1 + 2x)^6(1 - x)^4$, find the coefficient of x^3

2

$$\begin{aligned}
 d) \quad (1+2x)^6(1-x)^4 &= \left(\sum_{r=0}^6 \binom{6}{r} (2x)^r \right) \left(\sum_{r=0}^4 \binom{4}{r} (-x)^r \right) \\
 &= \left(\sum_{r=0}^6 \binom{6}{r} (2x)^{6-r} \right) \left(\sum_{r=0}^4 \binom{4}{r} (-x)^{4-r} \right) \\
 &= \binom{6}{3} (2x)^3 \times \binom{4}{1} (-x)^1 + \binom{6}{4} (2x)^2 \times \binom{4}{2} (-x)^2 + \\
 &\quad \binom{6}{5} (2x)^1 \times \binom{4}{3} (-x)^3 + \binom{6}{6} (2x)^0 \times \binom{4}{4} (-x)^4 \\
 \text{coefficient of } x^3: & 8 \times \binom{6}{3} \times \binom{4}{1} - 4 \times \binom{6}{4} \times \binom{4}{2} + 2 \times \binom{6}{5} \times \binom{4}{3} - \binom{6}{6} \times \binom{4}{4} \\
 &= -12
 \end{aligned}$$

A23	ABBOTSLEIGH	☒	☐
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Find the coefficient of x^3 in the expansion of $(2+x)(1-x)^5$

3

$$\begin{aligned}
 (b) \quad (2+x)(1-x)^5 \\
 \text{coeff. of } x^3 \text{ comes from} \\
 2 \times {}^5C_3 (-x)^3 + x \times {}^5C_2 (-x)^2 \\
 \text{coefficient} = 2 \times {}^5C_3 \times (-1)^3 + {}^5C_2 \times 1 \\
 = {}^5C_2 (-2 + 1) \\
 = \frac{5 \times 4}{2 \times 1} \times -1 \\
 = -10
 \end{aligned}$$

A24	BAULKHAM	☒	☐
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If the 3rd and 4th terms of the binomial $(1 + ax)^n$ are $264x^2$ and $1760x^3$ and $n > a$, find the values of a and n .

3

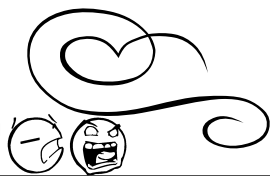
$$\begin{aligned}
 7c) \quad {}^nC_2 a^2 &= 264 \quad \therefore a = 1, 2, 4, 5, 10, \dots \\
 \frac{n(n-1)}{2} a^2 &= 264 \quad \text{--- (1)} \quad p \quad n = 22, 12, 7, 6, 4, 1 \\
 \text{but } n > a \quad \therefore \\
 {}^nC_3 a^3 &= 1760 \quad \text{trial } a = 1, 2, 4, 5. \\
 \therefore \frac{n(n-1)(n-2)}{6} a^3 &= 1760 \quad \text{Sub } a = 1 \quad n = 22 \text{ into (1)} \\
 \text{Sub (1) into (2)} \quad \frac{22(21)}{2} \cdot 1 &\neq 264 \\
 26 + \frac{(n-2)}{3} a &= 1760 \quad a = 2 \quad n = 12 \\
 88(n-2) a &= 1760 \quad \frac{12(11)}{2} \cdot 2^2 = 264 \checkmark \\
 (n-2) a &= 20 \quad \therefore a = 2 \quad n = 12 \quad (1) \\
 (n-2) &= \frac{20}{a} \quad \therefore a = 2 \quad n = 12 \quad (1) \\
 \text{now } n \text{ is a pos. integer}
 \end{aligned}$$

A25	CHERRYBROOK	☒	☐
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Find the coefficient of the term x^{-4} in the expansion of $\left(2x - \frac{1}{3x^2}\right)^8$

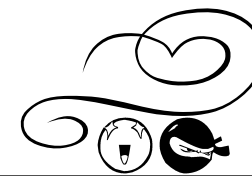
3

$$\begin{aligned}
 (c) \quad T_{k+1} &= {}^nC_k a^{n-k} b^k \\
 \text{So } T_{k+1} &= {}^8C_k (2x)^{8-k} \left(\frac{-1}{3x^2}\right)^k \\
 &= {}^8C_k 2^{8-k} x^{8-k} \left(\frac{-1}{3}\right)^k x^{-2k} \\
 &= {}^8C_k 2^{8-k} \left(\frac{-1}{3}\right)^k x^{8-k-2k} \\
 \therefore T_{k+1} &= K x^{8-3k} \\
 p \quad \text{So } x^{8-3k} &= x^{-4} \\
 \therefore k &= 4 \quad \leftarrow (1) \\
 \therefore \text{Coefficient is } & 8-4 = 4 \\
 T_{k+1} &= {}^8C_4 a^4 b^4 \\
 &= 70 (16x^4) \frac{1}{81x^8} \\
 &= \frac{1120}{81} x^{-4} \\
 &= 13 \frac{67}{81} \quad \leftarrow (1)
 \end{aligned}$$



Binomials

Part A: Terms



A26	CSSA	☒	☐
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In the expansion of $(1 - kx)^9$ the coefficient of x^6 is half that of the coefficient of x^5 . 3

Find the value of the constant k .

Consider terms of the expansion of $(1 - kx)^9$

$$\begin{aligned} \text{Term in } x^5 &= {}^9C_5(-kx)^5 & \text{Term in } x^6 &= {}^9C_6(-kx)^6 \\ &= -{}^9C_5k^5x^5 & &= {}^9C_6k^6x^6 \end{aligned}$$

$$\therefore -{}^9C_5k^5 = 2 \cdot {}^9C_6k^6$$

$$\begin{aligned} k &= -\frac{{}^9C_5}{{}^9C_6} \\ &= -\frac{1}{2} \times \frac{9!}{5!4!} \times \frac{3!6!}{9!} \\ &= -\frac{3}{4} \end{aligned}$$

A27	PENRITH	☒	☐
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In the expansion of $(4 + 2x - 3x^2)\left(2 - \frac{x}{5}\right)^6$, find the coefficient of x^5 3

$$\begin{aligned} &(4 + 2x - 3x^2)\left(2 - \frac{x}{5}\right)^6 \\ \text{General term: } &{}^6C_k 2^{6-k} (-1)^k \left(\frac{x}{5}\right)^k \\ \text{for } \left(2 - \frac{x}{5}\right)^6 &{}^6C_k 2^{6-k} 5^{-k} (-1)^k x^k \end{aligned}$$

when $k=5$, multiply by 4 (other brackets)

$$-{}^6C_5 2 \cdot 5^{-5} (4) = \frac{-48}{3125}$$

when $k=4$, multiply by 2 (other brackets)

$${}^6C_4 2^2 5^{-4} (2) = \frac{120}{625}$$

when $k=3$, multiply by (-3) (other brackets)

$$-{}^6C_3 2^3 5^{-3} (-3) = \frac{480}{125}$$

$$\frac{-48}{3125} + \frac{120}{625} + \frac{480}{125} = \frac{12552}{3125} \text{ or } 4.01664$$

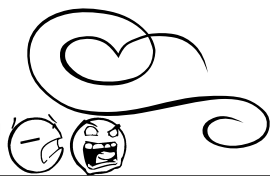
A28	ABBOTSLEIGH	☒	☐
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In the expansion of $(1 + ax)^n$ in ascending powers of x , the first three terms are 3

$1, -45x$ and $900x^2$. Find the values of a and n .

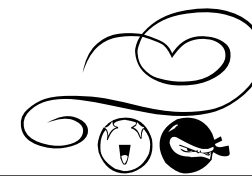
$$\begin{aligned} (1 + ax)^n &= 1 + \binom{n}{1}(ax) + \binom{n}{2}(ax)^2 + \dots \\ \therefore \binom{n}{1}a &= -45 \quad \text{and} \quad \binom{n}{2}a^2 = 900 \\ na &= -45 & n(n-1)a^2 &= 900 \\ a &= \frac{-45}{n} \quad \xrightarrow{\text{sub}} & \therefore n \frac{n-1}{2} \left(\frac{-45}{n}\right)^2 &= 900 \end{aligned}$$

$$\begin{aligned} \frac{n^2 - n}{n^2} &= \frac{900 \times 2}{45 \times 45} = \frac{8}{9} \\ 9n^2 - 9n &= 8n^2 \\ n^2 - 9n &= 0 \\ n(n-9) &= 0 \\ \therefore n &= 9 \quad (\text{reject } n=0) \\ a &= \frac{-45}{n} = \frac{-45}{9} = -5 \\ \therefore a &= -5, \quad n = 9. \end{aligned}$$



Binomials

Part B: Proving



B1	BAULKHAM HILLS	☒	☐
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By considering the binomial expansion of $(1+x)^{100} - (1-x)^{100}$, what is the value of

$$\binom{100}{3} + \binom{100}{5} + \dots + \binom{100}{99}?$$

1

(A) $2^{100} - 100$

(B) $2^{99} - 100$

10. B -

$$\begin{aligned} (1+x)^{100} - (1-x)^{100} &= \left[\binom{100}{0} + \binom{100}{1}x + \binom{100}{2}x^2 + \binom{100}{3}x^3 + \dots + \binom{100}{99}x^{99} + \binom{100}{100}x^{100} \right] \\ &\quad - \left[\binom{100}{0} - \binom{100}{1}x + \binom{100}{2}x^2 - \binom{100}{3}x^3 + \dots - \binom{100}{99}x^{99} + \binom{100}{100}x^{100} \right] \\ &= 2\left[\binom{100}{1}x + \binom{100}{3}x^3 + \dots + \binom{100}{99}x^{99} \right] \end{aligned}$$

Let $x=1$:

$$\begin{aligned} (1+1)^{100} - (1-1)^{100} &= 2\left[\binom{100}{1} + \binom{100}{3} + \dots + \binom{100}{99} \right] \\ 2^{100} - 0 &= 2\left[\binom{100}{1} + \binom{100}{3} + \dots + \binom{100}{99} \right] \\ 2^{100} - 2 &= 2\left[\binom{100}{1} + \binom{100}{3} + \dots + \binom{100}{99} \right] \\ 2^{100} - 100 &= \binom{100}{3} + \binom{100}{5} + \dots + \binom{100}{99} \end{aligned}$$

B2	BAULKHAM HILLS	☒	☐
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Using the fact that $(1+x)^4(1+x)^9 \equiv (1+x)^{13}$, show that

$$\binom{4}{0}\binom{9}{4} + \binom{4}{1}\binom{9}{3} + \binom{4}{2}\binom{9}{2} + \binom{4}{3}\binom{9}{1} + \binom{4}{4}\binom{9}{0} = \binom{13}{4}$$

2

14(d) By equating the coefficients of x^4 on both sides of the identity

$$\text{LHS: } (1+x)^4(1+x)^9 = \left[\binom{4}{0} + \binom{4}{1}x + \dots + \binom{4}{4}x^4 \right] \left[\binom{9}{0} + \binom{9}{1}x + \dots + \binom{9}{9}x^9 \right]$$

$$\text{Term with } x^4 = \binom{4}{0} \times \binom{9}{4}x^4 + \binom{4}{1}x \times \binom{9}{3}x^3 + \binom{4}{2}x^2 \times \binom{9}{2}x^2 + \binom{4}{3}x^3 \times \binom{9}{1}x + \binom{4}{4}x^4 \times \binom{9}{0}$$

$$= \left[\binom{4}{0}\binom{9}{4} + \binom{4}{1}\binom{9}{3} + \binom{4}{2}\binom{9}{2} + \binom{4}{3}\binom{9}{1} + \binom{4}{4}\binom{9}{0} \right] x^4$$

$$\text{RHS: } (1+x)^{13} = \binom{13}{0} + \binom{13}{1}x + \dots + \binom{13}{4}x^4 + \dots + \binom{13}{13}x^{13}$$

$$\text{Term with } x^4 = \binom{13}{4}$$

Since $(1+x)^4(1+x)^9 \equiv (1+x)^{13}$ the coefficients of x^4 must be equal

$$\text{i.e. } \binom{4}{0}\binom{9}{4} + \binom{4}{1}\binom{9}{3} + \binom{4}{2}\binom{9}{2} + \binom{4}{3}\binom{9}{1} + \binom{4}{4}\binom{9}{0} = \binom{13}{4}$$

B3	BAULKHAM HILLS	☒	☐
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Consider the expression:

4

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n$$

Multiplying by x , another identity is obtained:

$$x(1+x)^n = \binom{n}{0}x + \binom{n}{1}x^2 + \binom{n}{2}x^3 + \binom{n}{3}x^4 + \dots + \binom{n}{n-1}x^n + \binom{n}{n}x^{n+1}$$

Find the function $F(n)$ such that $2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1} = (n+2)F(n)$

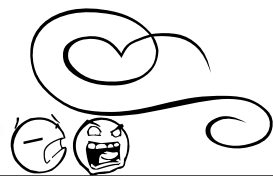
e) Differentiating both sides of the 2nd identity:

(*) correct solution
(3) correctly substitutes $x=1$ into 2 correct expressions
(2) correctly differentiates both sides
(1) correctly differentiates one side

$$\begin{aligned} \text{LHS} &= x \cdot n(1+x)^{n-1} + (1+x)^n \cdot 1 \\ &= (1+x)^{n-1} (nx + 1+x) \\ \text{RHS} &= 1 + 2\binom{n}{1}x + 3\binom{n}{2}x^2 + 4\binom{n}{3}x^3 + \dots + n\binom{n}{n-1}x^{n-1} \\ &\quad + (n+1)\binom{n}{n}x^n \end{aligned}$$

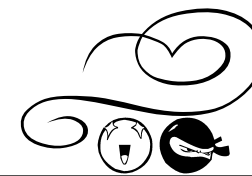
Substitute $x=1$ into both sides:

$$\begin{aligned} (1+1)^{n-1} (n+1+1) &= 1 + 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1} \\ &\quad + (n+1) \cdot 1 \\ 2^{n-1} (n+2) &= 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1} \\ &\quad + n+2 \\ \therefore 2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + n\binom{n}{n-1} &= 2^{n-1} (n+2) - (n+2) \\ &= (n+2)(2^{n-1} - 1) \quad \therefore F(x) = 2^{n-1} - 1 \end{aligned}$$



Binomials

Part B: Proving



B4	GOSFORD	☒	☐
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- (a) Write down the binomial expansion of $(1+x)^n$. 1
- (b) Hence show that ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1$. 2

(a) Write down the binomial expansion of $(1+x)^n$.

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n$$

(b) Hence show that ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1$.

substituting $x=1$, $2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

now ${}^nC_0 = 1 \therefore 2^n - 1 = {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

B5	INDEPENDENT	☒	☐
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- (i) Write down the binomial expansion of $(1+x)^n$. 1
- (ii) Hence show that ${}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = {}^nC_2 + {}^nC_4 + {}^nC_6 + \dots$. 2

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n$

d.ii. Outcomes assessed: ME11-5

Substitute $x=-1$. $0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 - {}^nC_5 + {}^nC_6 - \dots + {}^nC_n(-1)^n$

$\therefore {}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = {}^nC_2 + {}^nC_4 + {}^nC_6 + \dots$

B6	INDEPENDENT	☒	☐
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- (i) Write down the binomial expansion of $(1+x)^n$. 1
- (ii) Hence show that ${}^nC_1 + 2{}^nC_2 + 3{}^nC_3 + \dots + n{}^nC_n = n2^{n-1}$. 2

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n$

e.ii. Outcomes assessed: ME11-5

Differentiating both sides with respect to x : $n(1+x)^{n-1} = {}^nC_1 + {}^nC_2 \cdot 2x + {}^nC_3 \cdot 3x^2 + \dots + {}^nC_n \cdot nx^{n-1}$

Substituting $x=1$: $n2^{n-1} = {}^nC_1 + 2{}^nC_2 + 3{}^nC_3 + \dots + n{}^nC_n$

B7	NEAP	☒	☐
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- By considering the binomial expansion of $(1+x)^{10}$, what is the value of 1
- $$3\binom{10}{1} + 3^2\binom{10}{2} + 3^3\binom{10}{3} + \dots + 3^{10}\binom{10}{10}?$$
- A. 2^{10} B. 4^{10} C. $2^{10} - 1$ D. $4^{10} - 1$

B8	NEAP	☒	☐
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- By expanding both sides of the identity $(1+x)^{15} = (1+x)^{12}(1+x)^3$, prove that 3

$$\binom{15}{4} = \binom{12}{4} + 3\binom{12}{3} + 3\binom{12}{2} + \binom{12}{1}$$

B8:

$$(1+x)^{15} = (1+x)^{12}(1+x)^3$$

$$= (1+x)^{12}(1+3x+3x^2+x^3)$$

Since $\binom{15}{4}$ compare Coeff. of x^4 from B.S.

LHS: ${}^{15}C_4(1)^{11}x^4$ RHS: $(1+x)^{12} = (1+3x+3x^2+x^3)$

${}^{12}C_4$	x^4	1	$= {}^{12}C_4$
${}^{12}C_3$	x^3	$3x$	$= 3{}^{12}C_3$
${}^{12}C_2$	x^2	$3x^2$	$= 3{}^{12}C_2$
${}^{12}C_1$	x	x^3	$= {}^{12}C_1$

Adding $T_{k+1} = {}^{12}C_k 1^{12-k} x^k$ For $x^4: k=1$
 $= {}^{12}C_k x^k$ $x^4: k=2$
 $x^4: k=3$
 $x^4: k=4$

$\therefore LHS = RHS$

B9	PEM	☒	☐
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- Given $(1+x)^{n+3} = (1+x)^3(1+x)^n$, show $\binom{n+3}{3} = 1 + 3\binom{n}{1} + 3\binom{n}{2} + \binom{n}{3}$. 3

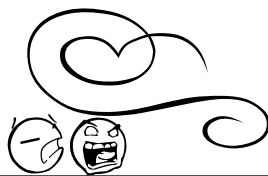
$(1+x)^{n+3} = (1+x)^3(1+x)^n$

$$= (1+3x+3x^2+x^3)\left(1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots\right)$$

coefficient of x^3 on LHS $= 1 + 3\binom{n}{2} + 3\binom{n}{1} + 1 \times \binom{n}{3}$

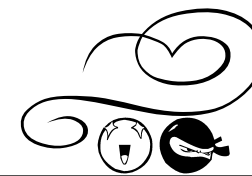
equating coefficients $\binom{n+3}{3} = 1 + 3\binom{n}{1} + 3\binom{n}{2} + \binom{n}{3}$

coefficient of x^3 on LHS $= \binom{n+3}{3}$



Binomials

Part B: Proving



B10	SCOTS	✓	✗
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- (i) Using the binomial theorem, obtain an expression for $(1+x)^{2n} + (1-x)^{2n}$. 2
- (ii) Hence, show that $1 + \binom{20}{2} + \binom{20}{4} + \dots + \binom{20}{20} = 2^{19}$. 1

B11	SYDNEY GIRLS	✓	✗
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By substituting a suitable value for x in the expansion $(1+x)^n$, show that

2

$$1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1} = 3^n - 2^n$$

b) $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n$
 Subst $x=2$
 $3^n = \binom{n}{0} + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1} + 2^n$
 $3^n - 2^n = \binom{n}{0} + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1}$
 proven.

B12	ACE	✓	✗
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- (i) Expand $\left(x + \frac{1}{x}\right)^5$ in descending powers of x . 1
- (iii) If $x + \frac{1}{x} = n$, express $x^5 + \frac{1}{x^5}$ in terms of n 3

14(d) (i)	$\left(x + \frac{1}{x}\right)^5 = x^5 + 5x^4\left(\frac{1}{x}\right) + 10x^3\left(\frac{1}{x}\right)^2 + 10x^2\left(\frac{1}{x}\right)^3 + 5x\left(\frac{1}{x}\right)^4 + \left(\frac{1}{x}\right)^5$ $= x^5 + 5x^3 + 10x + 10\left(\frac{1}{x}\right) + 5\left(\frac{1}{x}\right)^3 + \left(\frac{1}{x}\right)^5$	1 Mark: Correct answer.
14(d) (ii)	$\left(x + \frac{1}{x}\right)^5 = x^5 + \left(\frac{1}{x}\right)^5 + 5\left[x^3 + \left(\frac{1}{x}\right)^3\right] + 10\left(x + \frac{1}{x}\right)$ Now $x^3 + \left(\frac{1}{x}\right)^3 = \left(x + \frac{1}{x}\right)\left(x^2 - 1 + \left(\frac{1}{x}\right)^2\right)$ $= \left(x + \frac{1}{x}\right)\left(\left(x + \frac{1}{x}\right)^2 - 3\right)$ If $x + \frac{1}{x} = n$ $\left(x + \frac{1}{x}\right)^5 = x^5 + \left(\frac{1}{x}\right)^5 + 5\left[n^3 - n\right] + 10n$ $n^5 = x^5 + \left(\frac{1}{x}\right)^5 + 5[n^3 - n] + 10n$ $x^5 + \left(\frac{1}{x}\right)^5 = n^5 - 5n^3 + 5n$	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the solution. 1 Mark: Groups in pairs $x^n + \left(\frac{1}{x}\right)^n$